U1
AT32F405RCT7-7
X1
12MHz
C12
33pF
C11
33pF
R6
0Ω
VDDA
L2 600Ω
VCC
R13
1kΩ
R12
10kΩ
ATBOOT
SW4
R51
0Ω
C9
10uF
C5
100nF
C6
100nF
C7
100nF
C8
100nF
LDO3V
U7
AMS1117-3.3
L1
600Ω
R1
0Ω
C5V
TP2
C1
22uF
GND
R2
0Ω
TP1
C3
22uF
USB1
TYPE-C16PIN
R3
5.1kΩ
R4
5.1kΩ
USB_P
USB_M
R48
10kΩ
R49
10kΩ
R20
470Ω
LED2PB0
KT-0603R
R50
1kΩ
D24
U5
TVS5V
SW1
AT_RESET
C10
100nF
RESET
R7
12kΩ
U6
TVS5V
LAD1
LAD2
LAD3
R9 1kΩ
R52 1kΩ
R53 1kΩ
D25
D26
D27
R5
1kΩ
LED1
KT-0603R
HDR3_JT
JTAG
VTREF
SWDIO
SWCLK
SWO
RESET
JTAG
HDR1_MCU
LINK
HDR3_MCU
LINK
HDR2_MCU
MCU
FUNCPin
SWD_CLKPA.1
SWD_DIOPA.0
SWD_RSTPA.4
CDC_TXDPA.2
CDC_RXDPA.3
JTAG_TDIPA.7
JTAG_TDOPA.6
pc 0 TO 15 GOTO RG0..15
TITLE: Sheet_1
REV: 1.2
Company: ZOLEX
Sheet: 1/1
Date: 2026-02-11
Drawn By: AJR
EasyEDA
JOIN POWER RAILS
C18 10uF
C17 10uF
C13 10uF
C14 10uF
C15 10uF
C16 10uF
CRYSTAL
X101
12MHz
R11
1kΩ
C133
33pF
C134
33pF
RPXIN
RPXOUT
RPVCC
RP1V1
C117
2.2uF
C118
2.2uF
C112 10uF
C124 100nF
C123 100nF
C126 100nF
CD2 100nF
CD1 100nF
U101
RP2040
R14
33Ω
R15 33Ω
RPUSBDP
RPUSBDM
MCU INTER CONNECTS 1/2
RPSWD
RPSWCLK
RPTX0
RPRX0
RUN
LEDS_SCK
LEDS_DATA
R38
R39
RPGPIO0 NT16 PC0
RPGPIO1 NT15 PC1
RPGPIO2 NT14 PC2
RPGPIO3 NT13 PC3
RPGPIO4 NT9 PC4
RPGPIO5 NT10 PC5
RPGPIO6 NT11 PC6
RPGPIO7 NT12 PC7
RPGPIO8 NT22 PC8
RPGPIO9 NT23 PC9
RPGPIO10 NT25 PC10
RPGPIO11 NT24 PC11
RPGPIO12 NT29 PC12
RPGPIO13 NT28 PC13
RPGPIO14 NT26 PC14
RPGPIO15 NT27 PC15
Raspberry Pi
RP2040
MCU INTER CONNECTS 2/2
R21
470Ω
LD1
KT-0603R
RESET/RUN
R136
12kΩ
R138
1kΩ
100nF
C114
SW3
RP_RESET
SW2
boot
SPI FLASH
W25Q128JVSIQTR
SPI_FLASH1
QSPI_SS
QSPI_SD0
QSPI_SD1
QSPI_SD2
QSPI_SD3
QSPI_SCLK
SW5
BTN1
R54
1kΩ
PB10
FUNCPin
SWD_CLKPA.1
SWD_DIOPA.0
SWD_RSTPA.4
CDC_TXDPA.2
CDC_RXDPA.3
JTAG_TDIPA.7
JTAG_TDOPA.6
USB
USB2
TYPE-C16PIN
R17
5.1kΩ
R16
5.1kΩ
RPUSB5V
U9
TVS5V
U8
TVS5V
SPIxSCK OptionsMOSI Options
SPI1PA5, PB3PA7, PB5
SPI2PB13, PC7PB15, PC3
SPI3PB3, PC10PB5, PC12
HDR101_MCU
LINK
R18
R19
C2
100nF
R8
10kΩ
R10
1kΩ
C19
100nF
C20
100nF
C21
100nF
U22
74HC595D.
U2
74HC595D.
U4
74HC595D.
LEDS_RCK
D0
D1
D2
D3
D4
D5
D6
D7
D8
D9
D10
D11
D12
D13
D14
D15
D16
D17
D18
D19
D20
D21
D22
D23
R22
R23
R24
R25
R26
R27
R28
R29
R30
R37
R36
R35
R34
R33
R32
R31
R47
R40
R41
R42
R43
R44
R45
R46